

The damping of sound by wall turbulent shear layers

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(Received 22 August 1994; revised 22 March 1995; accepted 30 March 1995)

A frequency-dependent model is proposed for the eddy viscosity controlling the momentum and thermal wall boundary layers formed when an acoustic wave interacts with a low Mach number turbulent boundary layer whose thickness is much smaller than the acoustic wavelength. This is used to determine the effective acoustic admittance of the boundary layer, and to predict the attenuation of sound reflecting from a plane wall boundary layer, and low-frequency sound propagating in fully developed turbulent pipe flow. The predictions are shown to compare well with a recent experimental study of acoustic attenuation in pipe flows. © 1995 Acoustical Society of America.

PACS numbers: 43.28.Py, 43.28.Ra

INTRODUCTION

The two principal mechanisms leading to the attenuation of sound by turbulence are *scattering* (the production of secondary waves that radiate out of the line of propagation of the incident sound) and *absorption* (the transfer of energy to the turbulence).¹⁻⁸ Absorption tends to be more important at lower frequencies, when the periodic straining of vortex lines by the sound becomes irreversible, since there is then ample time in an acoustic cycle for energy to flow between the different degrees of freedom of the turbulence. This “scrambling” of acoustically strained vorticity is increased in a turbulent boundary layer, where gradients of the acoustic particle velocity are large. The analysis of this mechanism is made difficult by the absence of any detailed theoretical understanding of boundary layer turbulence, especially at high Mach numbers. Some limited progress has been made both experimentally⁹⁻¹⁴ and theoretically.^{8,11,12,15} None of the theoretical work has been able to account satisfactorily for certain peculiarities of the experimental data, but the recent publication by Peters *et al.*^{9,14} of data derived from careful measurements of the attenuation of sound in turbulent pipe flow has prompted a reexamination of the earlier analytical work. This work took no account of the frequency dependence of the response of the turbulence to straining by the sound, and was therefore unable to predict the attenuation over the whole frequency range, including a transition region in which molecular and turbulent transport processes are of comparable importance.

In this paper a frequency-dependent model of the effective turbulent boundary layer eddy viscosity is proposed that successfully predicts the attenuation of sound interacting with low Mach number wall turbulence for frequencies where the acoustic wavelength is much larger than the boundary layer thickness. The model is developed by reference to attenuation data obtained by Peters^{9,14} and extrapolated to zero Mach number, and predictions are compared with measurements of the damping of sound propagating in low Mach number pipe flow. The present theory is strictly applicable at small Mach numbers, but encouraging agreement is obtained for Mach numbers as high as 0.1.

The theory of acoustic attenuation at a plane wall in the absence of mean flow is briefly reviewed in Sec. I. This is

generalized in Sec. II to incorporate the influence of the transport of heat and momentum by turbulent mixing in a low Mach number wall turbulent shear layer. Application is made in Sec. III to sound propagation in fully developed turbulent pipe flow.

I. SUMMARY OF RESULTS IN THE ABSENCE OF MEAN FLOW

An acoustic wave whose pressure is given by the real part of $p_I \exp[i(k_1 x_1 - k_2 x_2 - \omega t)]$, $p_I = \text{constant}$, is incident from $x_2 > 0$ on a smooth wall lying in the plane $x_2 = 0$ of the coordinate system (x_1, x_2, x_3) . We take $\omega > 0$, so that $k_2 = \kappa_0 \cos \Theta > 0$, where Θ is the angle of incidence (see Fig. 1) and $\sqrt{k_1^2 + k_2^2} = \omega/c_0 \equiv \kappa_0$, where c_0 is the speed of sound. The net acoustic pressure is

$$p = p_I \{ e^{-ik_2 x_2} + R e^{ik_2 x_2} \} e^{i(k_1 x_1 - \omega t)}, \quad (1)$$

where R is a suitable reflection coefficient. In ideal fluid, with no dissipation, the vanishing of the normal velocity v_2 at the wall would imply that $R = 1$.

In a real fluid, viscous and thermal boundary layers are formed at the wall, across which the velocity and temperature reduce to their respective values at the surface. The widths of these boundary layers are, respectively, $\delta_\nu \sim (\nu/\omega)^{1/2}$ and $\delta_\chi \sim (\chi/\omega)^{1/2}$, where ν and χ are the kinematic viscosity and thermometric conductivity of the fluid; for most liquids and gases δ_ν, δ_χ are usually small compared to the acoustic wavelength $2\pi/\kappa_0$, and this will be assumed to be the case below. Equations for the reflection coefficient, which is now a complex-valued quantity with $|R| < 1$, are obtained by first equating the normal component of the acoustic particle velocity $(-i/\rho_0 \omega) \partial p / \partial x_2$ [calculated from expression (1) for p , with ρ_0 being the mean fluid density] to the normal velocity v_δ , say, at $x_2 = \delta$, where δ may be regarded as the larger of δ_ν and δ_χ . Since $\kappa_0 \delta \ll 1$, this gives

$$\frac{-k_2 p_I}{\rho_0 \omega} \{ 1 - R \} e^{i(k_1 x_1 - \omega t)} = v_\delta. \quad (2)$$

To determine v_δ we temporarily denote by $p', \rho', \mathbf{v}'$ the perturbation pressure, density, and velocity in the *acoustic* field. These quantities satisfy the equations of motion of an ideal fluid in $x_2 > \delta$. However, these equations may be as-

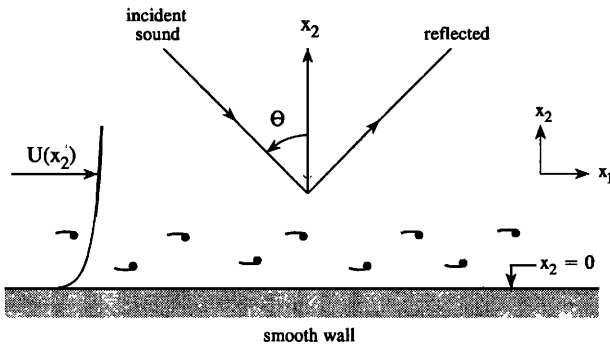


FIG. 1. The interaction of sound with a turbulent boundary layer.

summed to be satisfied by the acoustic variables within the whole of $x_2 > 0$, including the region occupied by the boundary layers. Since both the exact velocity and density ($\mathbf{v}$ and ρ , respectively) and the corresponding acoustic variables satisfy the linearized continuity equation, we can then form the difference

$$\frac{\partial(v_1 - v'_1)}{\partial x_1} + \frac{\partial(v_2 - v'_2)}{\partial x_2} + \frac{1}{\rho_0} \frac{\partial(\rho - \rho')}{\partial t} = 0,$$

and integrate over $0 < x_2 < \delta$. Because $v_2 \equiv v'_2$ at $x_2 = \delta$, $v_2 = 0$ at the rigid surface $x_2 = 0$, and $\kappa_0 \delta \ll 1$, we find

$$v_\delta = -\frac{\partial}{\partial x_1} \int_0^\infty (v_1 - v'_1) dx_2 - \frac{1}{\rho_0} \frac{\partial}{\partial t} \int_0^\infty (\rho - \rho') dx_2, \quad (3)$$

where the upper limits of integration have been extended to ∞ because the integrands vanish for $x_2 > \delta$.

The motion very close to the wall is in the tangential direction and is retarded by the viscous shear stress whose dominant component is $-\eta \partial v_1 / \partial x_1$, where η denotes the shear coefficient of viscosity. This retardation causes the fluid to be displaced outwards from the wall relative to the motion of an ideal fluid by an amount determined by the first integral in (3). The second integral represents the additional outward motion produced by the volumetric expansion (relative to an ideal fluid) that occurs when a temperature gradient $\partial T / \partial x_2$ is established in the boundary layer to bring the wall and fluid into equilibrium.

Near the wall and within the boundary layers, the pressure is approximately uniform and equal to

$$\bar{p} \equiv p_1 \{1 + R\} e^{i(k_1 x_1 - \omega t)},$$

and v_δ is usually expressed in terms of $\bar{p}$ and the acoustic admittance Y of the wall, by

$$Y \equiv Y_\nu + Y_\chi = -v_\delta / \bar{p}, \quad (4)$$

where, according to (3),

$$Y_\nu = \frac{1}{\bar{p}} \frac{\partial}{\partial x_1} \int_0^\infty (v_1 - v'_1) dx_2, \quad (5)$$

$$Y_\chi = \frac{1}{\rho_0 \bar{p}} \frac{\partial}{\partial t} \int_0^\infty (\rho - \rho') dx_2.$$

Y_ν and Y_χ are, respectively, the *shear stress* and *thermal* admittances of the wall.

When Y is known the reflection coefficient is given by

$$R = \frac{k_2 / \rho_0 \omega - Y}{k_2 / \rho_0 \omega + Y} = \frac{\cos \Theta - \rho_0 c_0 Y}{\cos \Theta + \rho_0 c_0 Y},$$

and the acoustic power Π dissipated per unit area of the wall is

$$\Pi = \frac{1}{2} |\bar{p}|^2 \text{Re}\{Y_\nu + Y_\chi\} = \frac{k_2 |p_1|^2}{2 \rho_0 \omega} (1 - |R|^2). \quad (6)$$

A. The shear stress admittance

The admittance Y is calculated by using the *boundary layer approximation* to the linearized equations of motion.¹⁶⁻¹⁹ For the tangential (x_1 -) momentum equation we have

$$\frac{\partial v_1}{\partial t} + \frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x_1} = \nu \frac{\partial^2 v_1}{\partial x_2^2}, \quad x_2 < \delta_\nu. \quad (7)$$

All field variables are evidently proportional to $e^{i(k_1 x_1 - \omega t)}$, so that the viscous term on the right is of order $1/(\kappa_0 \delta_\nu)^2 \gg 1$ relative to the omitted term $\nu \partial^2 v_1 / \partial x_1^2$; neglected compressive stresses of the type $\frac{1}{3} \nu \partial(\text{div } \mathbf{v}) / \partial x_1$ (including the contribution from the bulk viscosity) are smaller than the pressure gradient on the left-hand side by a factor of order $\kappa_0 \nu / c_0 \ll 1$. The left-hand side can be written $-i\omega(v_1 - \bar{v}_1)$, where $\bar{v}_1 = k_1 \bar{p} / \rho_0 \omega$ is the tangential acoustic particle velocity at the wall when the fluid is regarded as ideal. The Kirchhoff-Stokes solution of the resulting ordinary differential equation gives the velocity in the boundary layer in the form

$$v_1 = \bar{v}_1 \{1 - e^{i x_2 \sqrt{i \omega / \nu}}\}, \quad (8)$$

where the square root has positive imaginary part.

This formula yields the following equivalent expressions for shear stress admittance:

$$Y_\nu = \frac{k_1^2}{\rho_0 \omega} \sqrt{\frac{\nu}{i \omega}} \equiv \frac{k_1 \nu}{\omega \bar{p}} \left(\frac{\partial v_1}{\partial x_2} \right)_0, \quad (9)$$

where the derivative is evaluated at the wall. The contribution Π_ν to the dissipated sound power Π (6) is

$$\Pi_\nu = \frac{\sqrt{\nu / 2 \omega}}{2 \rho_0 \omega} \left| \frac{\partial \bar{p}}{\partial x_1} \right|^2 \equiv \left\langle \bar{v}_1 \eta \left(\frac{\partial v_1}{\partial x_2} \right)_0 \right\rangle, \quad (10)$$

where $\eta = \rho_0 \nu$ is the shear coefficient of viscosity, and the angle brackets denote a time average which is evaluated by taking the real parts of the complex representations of $\bar{v}_1$ and v_1 . Equation (10) can also be expressed in the form

$$\Pi_\nu = - \int_0^\infty \left\langle v_1 \frac{\partial \bar{p}}{\partial x_1} \right\rangle dx_2$$

$$= - \frac{1}{2} |\bar{v}_1|^2 \int_0^\infty \text{Im} \left\{ \frac{v_1(x_2)}{\bar{v}_1} \right\} dx_2. \quad (11)$$

The integrands vanish in the acoustic region $x_2 > \delta_\nu$, where v_1 is in quadrature with $\partial \bar{p} / \partial x_1$.

B. The thermal admittance

A similar calculation, involving the assumption that the heat capacity of the wall is sufficiently large that the fluctuating temperature T vanishes at $x_2=0$, gives

$$Y_\chi = \frac{\omega\beta}{\rho_0 c_p} \sqrt{\frac{\chi}{i\omega}} = \frac{\beta\chi}{\bar{p}} \left(\frac{\partial T}{\partial x_2} \right)_0, \quad (12)$$

where β is the coefficient of expansion at constant pressure, c_p the specific heat at constant pressure, and the final derivative is evaluated at the wall. The power dissipated by thermal action is $\Pi_\chi = \omega\beta\sqrt{\chi/2\omega}|\bar{p}|^2/2\rho_0 c_p$.

Combining Eqs. (9) and (12), the overall acoustic admittance becomes

$$Y = \frac{e^{-i\pi/4}}{\rho_0 \omega^{3/2}} \left[k_1^2 \sqrt{\nu} + \frac{\beta\omega^2}{c_p} \sqrt{\chi} \right], \quad (13)$$

and the acoustic power dissipated per unit area is

$$\Pi = \frac{\sqrt{\omega}}{2\sqrt{2}\rho_0 c_0^2} |\bar{p}|^2 \left[\sqrt{\nu} \sin^2 \Theta + \frac{\beta c_0^2}{c_p} \sqrt{\chi} \right]. \quad (14)$$

The Prandtl number $Pr = \nu/\chi$ can be large in liquids (≈ 6.7 in water), and is $O(1)$ in gases, but the ratio of the contributions to the dissipation from the shear stress and thermal terms is of order $\sqrt{Pr}$, which are therefore of comparable importance. In an ideal gas $\beta = 1/T_0$, and $\beta c_0^2/c_p = \gamma - 1$, where γ is the ratio of the specific heats and T_0 is the mean temperature.

II. DISSIPATION BY WALL TURBULENT SHEAR FLOWS

Consider turbulent boundary layer flow in $x_2 > 0$ over the smooth rigid wall $x_2 = 0$ (Fig. 1) with mean velocity $U(x_2)$ in the x_1 direction. The variation of U over distances comparable to the acoustic wavelength in the streamwise direction x_1 will be neglected (an excellent approximation in turbulent pipe flow, and for zero-pressure gradient flow over a flat wall²⁰), and the Mach number in the exterior mean flow will be assumed to be small enough that convection of sound can be ignored. The damping of sound may then be investigated by the method of Sec. I. In the absence of flow the transfer of momentum and heat across the wall boundary layers is controlled by molecular diffusion; this mechanism is negligible in the turbulent region, where diffusion is caused by the *mixing* of fluid particles.

The generalized form of the momentum boundary layer equation (7) is formed by taking a suitable average of the nonlinear momentum equation. When this equation is expressed in Reynolds' form,¹⁸ the term responsible for the transfer of x_1 momentum away from the wall is the negative Reynolds stress $\rho v_1 v_2 \approx \rho_0 v_1 v_2$. For the acoustic field (1) the *mean* rate of transfer of tangential acoustic momentum is equal to the difference

$$\overline{\rho_0 v_1 v_2} - \rho_0 \langle v_1 v_2 \rangle$$

between the ensemble average negative Reynolds stress $\rho_0 v_1 v_2$ in the presence of the sound and that, $\rho_0 \langle v_1 v_2 \rangle$, of the background turbulent motion (no sound). The boundary

layer approximation for the acoustic particle velocity $v_1(x_1)$ is therefore

$$\frac{\partial v_1}{\partial t} + \frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x_1} = \frac{-\partial}{\partial x_2} (\overline{v_1 v_2} - \langle v_1 v_2 \rangle) + \nu \frac{\partial^2 v_1}{\partial x_2^2}, \quad (15)$$

where all of the velocities on the right-hand side may be regarded as relative to the local mean flow. The convection term

$$U(x_2) \frac{\partial v_1}{\partial x_1} = ik_1 U(x_2) v_1 \approx O(M) \frac{\partial v_1}{\partial t}, \quad M = \frac{U}{c_0},$$

is small when $M \ll 1$, and has been discarded.

There is no satisfactory deductive theory for the fluctuating Reynolds stress in a turbulent boundary layer, even when the flow is not modulated by incident sound. Recent developments do not significantly advance the original phenomenological treatment due to Prandtl,²⁰ on which the following discussion will be based, and which has the advantage of yielding an analytical solution of the boundary layer equation.

In equilibrium boundary layer flow (no sound) Prandtl equated the shear stress $-\rho_0 \langle v_1 v_2 \rangle$ (which is the rate of diffusion of x_1 momentum into the wall, and must be *positive* if the mean drag is in the x_1 direction) at distance x_2 from the wall to $\rho_0 l_p^2 (\partial U(x_2)/\partial x_2)^2$, where l_p is a *mixing length* over which distance a fluid particle is convected before being thoroughly mixed with its turbulent environment. This is equivalent to writing $-\rho_0 \langle v_1 v_2 \rangle = \rho_0 \epsilon_m \partial U(x_2)/\partial x_2$, where $\epsilon_m \equiv l_p^2 \partial U(x_2)/\partial x_2$ is an *eddy viscosity*. Experiments at low Mach numbers reveal that the mean shear stress is practically constant and equal to $\rho_0 v_*^2$ over most of the boundary layer (v_* being the friction velocity^{16,20}), so that $\epsilon_m \partial U/\partial x_2 = v_*^2$ within the turbulent region. By making the additional hypothesis that $\epsilon_m = \kappa v_* x_2$, where κ is the *von Karman* constant, Prandtl deduced the following logarithmic formula for the mean velocity profile:

$$\frac{U(x_2)}{v_*} = \frac{1}{\kappa} \ln \left(\frac{x_2 v_*}{\nu} \right) + \text{constant}, \quad l_p = \kappa x_2.$$

This agrees with boundary layer measurements over an extensive region of the boundary layer from about $x_2 v_*/\nu = 30$ to 2×10^3 , provided $\kappa \approx 0.41$ and the constant ≈ 5 .^{16,20} In this approximation the mixing length l_p is proportional to the distance from the wall, but it must actually tend to zero more rapidly than this within the "viscous sublayer" ($x_2 v_*/\nu < 7$), where the Reynolds stress is negligible. The sublayer motion is dominated by molecular diffusion, i.e., $\nu \partial U/\partial x_2 = v_*^2$, where $U(x_2)$ varies linearly: $U(x_2)/v_* \approx x_2 v_*/\nu$.

These ideas can be used to model the interaction of the tangential acoustic particle velocity v_1 with the boundary layer by writing

$$\overline{v_1 v_2} = - \left\{ l_p \frac{\partial U}{\partial x_2} + l'_p \frac{\partial v_1}{\partial x_2} \right\}^2.$$

The linearized boundary layer equation (15) then becomes

$$\frac{\partial v_1}{\partial t} + \frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x_1} = \frac{\partial}{\partial x_2} \left[(\nu + 2\epsilon_m(x_2)) \frac{\partial v_1}{\partial x_2} \right], \quad (16)$$

where $\epsilon_m(x_2) \equiv l_P l'_P \partial U / \partial x_2$ will generally depend on the frequency of the sound.

At low frequencies the acoustic boundary layer defined by the solution of this equation must extend far out into the turbulent flow; the disturbed motion should be approximately quasistatic with $\epsilon_m \approx \kappa v_* x_2$ over most of the boundary layer. The turbulence relaxation time, during which the turbulence shear stress evolves to accommodate the change $v_1(x_2)$ in velocity profile, is then a small fraction of the acoustic period, and kinetic energy acquired by the turbulence when strained by the sound has ample time to flow between the different turbulence modes and is not, therefore, recovered by the acoustic field when the strain is released. In the opposite extreme of very high frequencies, the acoustic boundary layer is entirely within the viscous sublayer and the turbulence has no influence on acoustic dissipation.

There must exist, however, a range of intermediate frequencies for which the principal acoustic-turbulence interactions occur in a near wall region where both viscous and turbulence diffusion are significant, and where the simple linear model $\epsilon_m \approx \kappa v_* x_2$ is inadequate. This range is centered on a critical frequency $\omega_*^{9,13,14,21}$ whose value can be roughly estimated to correspond to the condition that the edge $x_2 \approx 7\nu/v_*$ of the viscous sublayer should fall in the middle of the region of maximum slope of the acoustic particle velocity profile. This occurs at about $x_2(\omega_*/\nu)^{1/2} \approx \frac{3}{4}$ when the profile is approximated by the Kirchhoff-Stokes formula (8), so that $\omega_* \nu/v_*^2 \approx 0.01$. Experiments¹³ also suggest that at this frequency the turbulence relaxation time near the wall is comparable to the acoustic period.

The following extension of Prandtl's linear approximation can be used to model both the frequency dependence of the acoustic-boundary layer interaction and the rapid decrease of ϵ_m to zero within the sublayer:

$$\begin{aligned} \epsilon_m &= 0, & x_2 < \delta_v(\omega), \\ &= \kappa v_* [x_2 - \delta_v(\omega)], & x_2 > \delta_v(\omega). \end{aligned} \quad (17)$$

The frequency-dependent length δ_v must be approximately equal to the sublayer thickness ($\approx 7\nu/v_*$) when $\omega/\omega_* \rightarrow 0$ and the interaction becomes quasistatic. At higher frequencies δ_v must increase to account for the increased importance of viscous diffusion, which is dominant for $\omega \gg \omega_*$, and for the reduced efficiency of turbulent diffusion that occurs when the acoustic period decreases below the turbulence relaxation time. The following empirical formula fulfills these requirements:

$$\begin{aligned} \frac{\delta_v v_*}{\nu} &= 6.5 \left(1 + \frac{\sigma(\omega/\omega_*)^3}{1 + (\omega/\omega_*)^3} \right), & \omega_* \nu/v_*^2 \approx 0.01, \\ &\omega > 0, \end{aligned} \quad (18)$$

where σ is a constant. The value $\delta_v v_*/\nu = 6.5$ (rather than 7) yields optimal agreement with experiment at low frequencies. The constant σ controls the acoustic damping at high frequencies, for which it is shown below that $\sigma \approx 1.7$ yields the best correspondence with experiment.

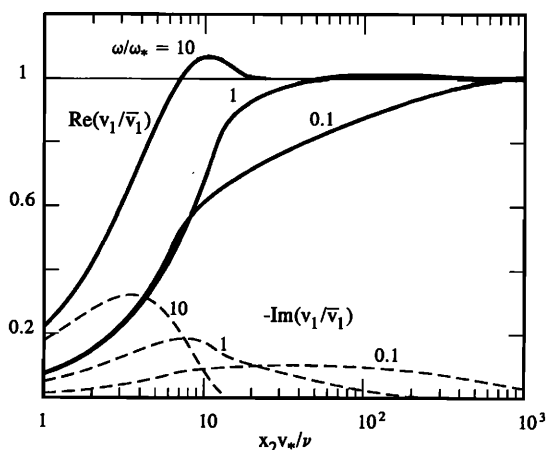


FIG. 2. The real (—) and imaginary (---) parts of the acoustic momentum boundary layer defined by Eq. (19) when $\sigma = 1.7$ for sound interacting with a turbulent boundary layer at low Mach numbers.

An additional modification to (17) must be made at frequencies that are so small that the region of maximum slope of the acoustic boundary layer velocity profile extends beyond about $x_2 v_*/\nu \approx 2 \times 10^3$, where Prandtl's linear model for ϵ_m is not valid.²⁰ This case will be discussed separately.

A. The boundary layer admittance

For sound of frequency ω the left-hand side of (16) is replaced by $-i\omega(v_1 - \bar{v}_1) \equiv -i\omega(v_1 - k_1 \bar{p}/\rho_0 \omega)$. When ϵ_m is defined by (17) and (18) the solution that satisfies the no-slip condition at the wall, and the continuity of v_1 and $[\nu + 2\epsilon_m(x_2)] \partial v_1 / \partial x_2$ at $x_2 = \delta_v$, is found to be

$$\begin{aligned} \frac{v_1(x_2)}{\bar{v}_1} &= 1 - \exp \left\{ ix_2 \sqrt{\frac{i\omega}{\nu}} \right\} \\ &+ \frac{H_1^{(1)}(a) \cos b + iH_0^{(1)}(a) \sin b}{H_0^{(1)}(a) \cos b + H_1^{(1)}(a) \sin b} \sin \left\{ x_2 \sqrt{\frac{i\omega}{\nu}} \right\}, \\ &0 < x_2 < \delta_v, \\ &= 1 - \frac{H_0^{(1)} \{ a [1 + (2\kappa v_*/\nu)(x_2 - \delta_v)]^{1/2} \}}{H_0^{(1)}(a) \cos b + H_1^{(1)}(a) \sin b}, \\ &x_2 > \delta_v, \end{aligned} \quad (19)$$

where $a = (i\omega \nu / \kappa^2 v_*^2)^{1/2}$, $b = \delta_v(\omega/\nu)^{1/2}$ (with positive imaginary parts), and $H_n^{(1)}$ is a Hankel function of order n .

The solution is illustrated in Fig. 2 for three representative values of ω/ω_* when $\sigma = 1.7$. When $\omega/\omega_* = 10$ the velocity profile is similar to that defined by the Kirchhoff-Stokes formula (8) and is controlled by viscous diffusion. The principal contribution to the dissipation integral (11) is from the neighborhood of $x_2 v_*/\nu \approx 3$, well within the viscous sublayer, where $-\text{Im}(v_1/\bar{v}_1)$ assumes its maximum value. At lower frequencies the maximum of $-\text{Im}(v_1/\bar{v}_1)$ is much broader, and the region of high dissipation can extend far out into the turbulent flow. When $\omega/\omega_* \ll 1$ the profile of $\text{Re}\{v_1/\bar{v}_1\}$ is similar to that of the mean velocity $U(x_2)$, increasing linearly out to about $x_2 v_*/\nu \approx 8$, and roughly logarithmically for larger values of $x_2 v_*/\nu$.²⁰

The definition (5) can be used to obtain the following generalization of the no-flow shear stress admittance (9), which is given by the same formula $Y_v = (k_1 \nu / \omega \bar{p}) (\partial v_1 / \partial x_2)_0$, with the explicit representation

$$Y_v = (k_1^2 / \rho_0 \omega) \sqrt{\nu / i \omega} F_A(\sqrt{i \omega \nu / \kappa^2 v_*^2}, \delta_v \sqrt{i \omega / \nu}), \quad (20)$$

where

$$F_A(a, b) = \frac{i[H_1^{(1)}(a) \cos b - H_0^{(1)}(a) \sin b]}{H_0^{(1)}(a) \cos b + H_1^{(1)}(a) \sin b}. \quad (21)$$

It is easily verified that $F_A \rightarrow 1$ as $v_* \rightarrow 0$, and that (20) then reduces to the no-flow formula (9).

The influence of turbulence mixing on the thermal admittance Y_χ is obtained by appealing to *Reynolds analogy* between the transport of momentum and heat by turbulent convection,²⁰ according to which the thermal boundary layer equation assumes the form

$$\frac{\partial}{\partial t} (T - \bar{T}) = \frac{\partial}{\partial x_2} \left([\chi + 2\epsilon_T(x_2)] \frac{\partial T}{\partial x_2} \right), \quad (22)$$

where $\bar{T} = \bar{p} / \rho_0 c_p$ is the adiabatic temperature fluctuation produced by the sound, $\epsilon_T(x_2)$ is an eddy diffusivity given by $\epsilon_T(x_2) = \epsilon_m(x_2) / P_t$, and $\epsilon_m(x_2)$ is defined as in (17) and (18) for time harmonic sound waves. Here, P_t is a turbulence *Prandtl number*, which is assumed to be constant; in air $P_t \approx 0.7$.^{20,22} In this case we find

$$Y_\chi = \frac{\beta \chi}{\bar{p}} \left(\frac{\partial T}{\partial x_2} \right)_0 = \frac{\omega \beta}{\rho_0 c_p} \sqrt{\frac{\chi}{i \omega}} F_A \left(\sqrt{\frac{i \omega \chi P_t^2}{\kappa^2 v_*^2}}, \delta_v \sqrt{\frac{i \omega}{\chi}} \right). \quad (23)$$

The net boundary layer admittance is therefore

$$Y = \frac{e^{-i\pi/4}}{\rho_0 \omega^{3/2}} \left[n_1^2 \sqrt{\nu} F_A \left(\sqrt{\frac{i \omega \nu}{\kappa^2 v_*^2}}, \delta_v \sqrt{\frac{i \omega}{\nu}} \right) + \frac{\beta \omega^2}{c_p} \sqrt{\chi} F_A \left(\sqrt{\frac{i \omega \chi P_t^2}{\kappa^2 v_*^2}}, \delta_v \sqrt{\frac{i \omega}{\chi}} \right) \right] \quad (24)$$

and the acoustic power dissipated per unit area of the wall is

$$\begin{aligned} \Pi = & \frac{\sqrt{\omega}}{2\sqrt{2}\rho_0 c_0^2} |\bar{p}|^2 \operatorname{Re} \left(\sqrt{2} e^{-i\pi/4} \left[\sqrt{\nu} \sin^2 \Theta \right. \right. \\ & \times F_A \left(\sqrt{\frac{i \omega \nu}{\kappa^2 v_*^2}}, \delta_v \sqrt{\frac{i \omega}{\nu}} \right) \\ & \left. \left. + \frac{\beta c_0^2}{c_p} \sqrt{\chi} F_A \left(\sqrt{\frac{i \omega \chi P_t^2}{\kappa^2 v_*^2}}, \delta_v \sqrt{\frac{i \omega}{\chi}} \right) \right] \right). \end{aligned} \quad (25)$$

A comparison of this formula with recent measurements of Peters *et al.*^{9,14} of the damping of axially propagating sound in a pipe containing low Mach number turbulent air flow ($\nu = 0.15$ cm/s, $\chi = 0.2$ cm/s, $P_t = 0.7$) can be used to estimate the value of the constant σ in Eq. (18). The acoustic power dissipated per unit area of the pipe wall is given by (25) with $\Theta = 90^\circ$, provided the acoustic boundary layer thickness is small compared to the pipe radius. It turns out

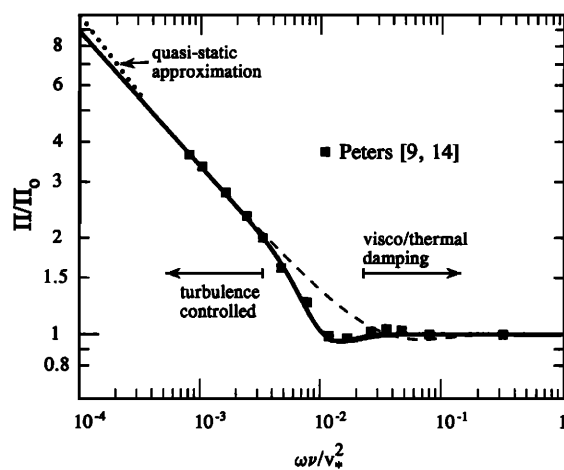


FIG. 3. Turbulent boundary layer damping of sound in air at very low Mach numbers relative to the no-flow attenuation: —, the prediction (25) of Π/Π_0 when δ_v is given by Eq. (18) with $\sigma = 1.7$, where Π_0 is the dissipation for no mean flow given by (14); ---, the corresponding prediction when $\delta_v v_* / \nu = 6.5$.

that the mean flow has a significant effect on the acoustic wave number and also, therefore, on the attenuation, even when the Mach number is small (see Sec. III). The experimental results shown in Fig. 3 represent extrapolations to zero Mach number obtained by averaging measurements for waves of the same frequency propagating in the upstream and downstream directions. The solid curve is the prediction (25) of Π/Π_0 when δ_v is given by (18) with $\sigma = 1.7$, where Π_0 is the dissipation for no mean flow given by (14).

This value of σ yields favorable overall agreement between theory and experiment, especially in the transition region centered on the critical frequency $\omega_* \nu / v_*^2 \approx 0.01$, where the attenuation falls below that in the absence of flow. In the turbulence controlled region, $\omega \ll \omega_*$, the thickness parameter $\delta_v v_* / \nu \rightarrow 6.5$. The dashed curve in the figure illustrates the corresponding prediction when δ_v has this constant value for all frequencies. The dissipation in the neighborhood of the critical frequency is now significantly overestimated, and the minimum in the attenuation is shifted to a much higher frequency. Similar discrepancies occur for other frequency-independent models of the eddy viscosity discussed in Refs. 8, 9, 12, 14, and 15.

The above theory and the experimental results shown in Fig. 3 are for sound propagating in planes parallel to the mean flow direction and the x_2 axis. According to Reynolds' analogy, however, attenuations of comparable magnitudes should be experienced by arbitrarily incident sound, although no experimental data are available to confirm this assertion.

B. Damping at very low frequencies

The turbulent flow should respond quasistatically to incident sound of very low frequency, i.e., the altered properties of the insonified boundary layer should be the same as for a boundary layer whose exterior mean flow velocity is $U_\infty + \bar{v}_1$ [$U_\infty = U(\infty)$]. When expressed as a function of $\bar{v}_1$, the dissipated power Π should therefore exhibit no dependence on frequency.

The dissipation Π_ν due to momentum transfer across the boundary layer is equal to $\langle \bar{v}_1 \eta (\partial v_1 / \partial x_2)_0 \rangle \equiv \langle \bar{v}_1 \Delta \tau_* \rangle$, where $\Delta \tau_*$ denotes the perturbation wall shear stress. For sound propagating in planes parallel to the mean flow and the x_2 axis, $\Delta \tau_* \approx 2\rho_0 v_* \Delta v_*$, where Δv_* is the quasistatic change in friction velocity produced by the sound. Δv_* can be expressed in terms of $\bar{v}_1$ by using the following equilibrium boundary layer equations:^{16,20}

$$U_\infty / v_* = (1/\kappa) \ln(\delta v_* / \nu) + 5, \quad (26)$$

$$U_\infty \delta / \nu = \{C U_\infty x_1 / \nu\}^{4/5}, \quad (27)$$

where δ is the turbulent boundary layer thickness. The distance x_1 is measured from an upstream origin roughly coinciding with the leading edge of the wall on which the boundary layer is formed, and C is a constant.

We calculate $\Delta v_* / v_*$ by making the substitutions

$$U_\infty \rightarrow U_\infty + \bar{v}_1, \quad \delta \rightarrow \delta + \Delta \delta, \quad v_* \rightarrow v_* + \Delta v_*$$

in (26) and (27) for fixed x_1 . By expanding to first order in perturbation quantities, it is then easily shown that

$$\Pi_\nu = \langle \bar{v}_1 \Delta \tau_* \rangle = \frac{v_*^2 |\bar{p}|^2 \sin^2 \Theta (\kappa + v_* / 5 U_\infty)}{\rho_0 c_0^2 U_\infty (\kappa + v_* / U_\infty)}. \quad (28)$$

The following argument gives the corresponding estimate for Π_χ . Equation (25) may be assumed to represent the attenuation at all frequencies, including the quasistatic limit, provided the correct functional form of F_A is used. As $\omega / \omega_* \rightarrow 0$, however, $\sqrt{\nu} F_A(\sqrt{i\omega\nu/\kappa^2 v_*^2}, \delta_v \sqrt{i\omega/\nu})$ cannot depend on the inner length scale δ_v nor on the kinematic viscosity ν , i.e.,

$$\sqrt{\nu} F_A(\sqrt{i\omega\nu/\kappa^2 v_*^2}, \delta_v \sqrt{i\omega/\nu}) \rightarrow C' \sqrt{v_*^2 / \omega} \quad \text{for } \omega \ll \omega_*,$$

where C' does not depend on ν or ω . A comparison of (28) with the value of Π_ν obtained by substituting this limiting value of F_A into the first term in the square brackets of (25) shows that $C' = 2(v_* / U_\infty)(\kappa + v_* / 5 U_\infty) / (\kappa + v_* / U_\infty)$. This can be used to find the corresponding limiting value of the second term in the square brackets of (25), from which it is deduced that, in the quasistatic approximation,

$$\Pi = \frac{v_*^2 |\bar{p}|^2 (\kappa + v_* / 5 U_\infty)}{\rho_0 c_0^2 U_\infty (\kappa + v_* / U_\infty)} \left[\sin^2 \Theta + \frac{\beta c_0^2}{P_t c_p} \right]. \quad (29)$$

This implies that Π/Π_0 increases like $(v_*^2 / \omega \nu)^{1/2}$ as $\omega \rightarrow 0$, whereas according to Eqs. (18) and (25), Π/Π_0 grows more slowly like $(v_*^2 / \omega \nu)^{1/2} / \ln\{(v_*^2 / \omega \nu)\}$. The value of v_* / U_∞ depends on the boundary layer Reynolds number $U_\infty \delta / \nu$, and is equal to about 0.035 when $U_\infty \delta / \nu \approx 10^5 - 10^6$.²⁰ With this value of v_* / U_∞ the quasistatic damping of the sound in Peters' experiment^{9,14} is shown by the dotted curve in Fig. 3.

III. SOUND PROPAGATION IN TURBULENT PIPE FLOW

Consider sound propagating in a nominally infinite smooth pipe of uniform cross section A and perimeter l , in the presence of fully developed, low Mach number turbulent

flow at mean velocity U in the direction of the x_1 axis. The acoustic frequency is assumed to be small enough that the sound, whose wavelength is then much larger than the pipe diameter, can propagate only one-dimensionally parallel to the axis. For a circular pipe of radius R this requires ω to be less than about $1.84 c_0 / R$.^{23,24}

At high Reynolds number the mean velocity varies slowly with position over the pipe cross section except near the wall. In these circumstances U can be regarded as constant in the core of the flow where, also, all dissipative mechanisms may be neglected in comparison with the damping caused by the acoustic-turbulence interactions at the wall. The axial momentum and continuity equations for the sound therefore assume the simplified forms

$$\frac{Dv_1}{Dt} + \frac{1}{\rho_0} \frac{\partial p}{\partial x_1} = 0, \quad \text{div } \mathbf{v} + \frac{1}{\rho_0 c_0^2} \frac{Dp}{Dt} = 0, \quad (30)$$

where $D/Dt \equiv \partial/\partial t + U \partial/\partial x_1$.

One-dimensional equations governing the propagation of sound are obtained by averaging equations (30) over the cross section of the core flow, where U is constant. At large Reynolds number this area is only slightly smaller than A , and the average acoustic particle velocity and pressure may therefore be defined by

$$\bar{v}_1(x_1, t) = \frac{1}{A} \int v_1 dS, \quad \bar{p}(x_1, t) = \frac{1}{A} \int p dS,$$

so that

$$\frac{D\bar{v}_1}{Dt} + \frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x_1} = 0, \quad \frac{\partial \bar{v}_1}{\partial x_1} + \frac{1}{\rho_0 c_0^2} \frac{D\bar{p}}{Dt} = -\frac{4}{D_p} v_\delta,$$

where $D_p = 4A/l$ is the hydraulic diameter of the pipe, and v_δ is the normal component of the acoustic particle velocity at the outer edge of the wall boundary layer (directed inwards from the wall). By eliminating $\bar{v}_1$ the pressure is found to satisfy

$$\frac{1}{c_0^2} \frac{D^2 \bar{p}}{Dt^2} - \frac{\partial^2 \bar{p}}{\partial x_1^2} = -\frac{4\rho_0}{D_p} \frac{Dv_\delta}{Dt},$$

where the term on the right accounts for the interaction of the sound with the turbulent wall flow.

Let the acoustic wave be given by the real part of $\bar{p} = p e^{i(kx_1 - \omega t)}$, $\omega > 0$, and set $v_\delta = -\bar{p} Y$, $Y(k, \omega)$ being the admittance of the wall shear layer. The wave equation yields the following dispersion relation between k and ω :

$$k^2 - (\kappa_0 - Mk)^2 = -\frac{4i\rho_0 c_0}{D_p} (\kappa_0 - Mk) Y(k, \omega), \quad (31)$$

where $M = U/c_0$. The admittance vanishes for a nonturbulent, ideal fluid, and the roots of this equation are then $k = \pm \kappa_0 / (1 \pm M)$.²⁵ The correction due to the wall shear layer is generally small, and is obtained by substituting

$$k = \pm \kappa_0 / (1 \pm M) \pm k'_\pm,$$

into Eq. (31), where $|k'_\pm| \ll \kappa_0$. To first order this gives

$$k'_\pm \approx \frac{2i\rho_0 c_0}{(1 \pm M) D_p} Y\left(\frac{\pm \kappa_0}{1 \pm M}, \omega\right). \quad (32)$$

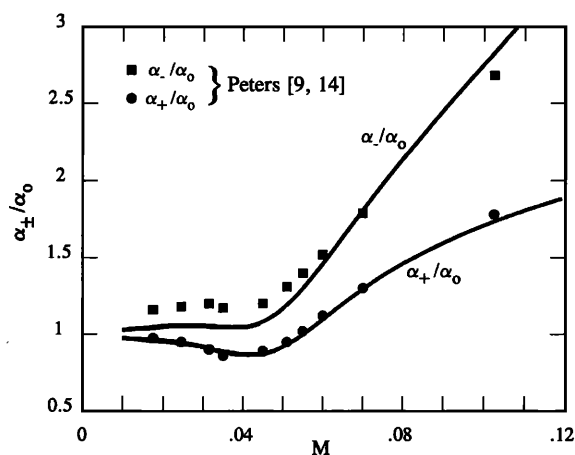


FIG. 4. Dependence on mean flow Mach number M of the damping coefficients $\alpha_{\pm}$ for plane waves propagating, respectively, in the upstream and downstream directions in a circular cylindrical pipe; α_0 is the damping coefficient in the absence of flow. The predictions (—) are based on Eqs. (33) and (34), with δ_v defined as in (18) with $\sigma=1.7$; the measurements are from Refs. 9 and 14.

The imaginary part of $k'_{\pm}$ determines the attenuation. In the absence of flow the dissipation occurs in viscous and thermal boundary layers of the type discussed in Sec. I. When the pipe radius is much larger than their respective thicknesses δ_v and δ_{χ} (i.e., when $\omega \gg \nu/D_p^2$), the plane wall approximation (13) for the admittance can be used to obtain

$$\text{Im}\{k'_{\pm}\} = \pm \alpha_0, \quad \text{where } \alpha_0 = \frac{\sqrt{2}\omega}{c_0 D_p} \left(\sqrt{\nu} + \frac{\beta c_0^2}{c_p} \sqrt{\chi} \right). \quad (33)$$

The damping in the presence of turbulent flow is determined by the admittance (24). Setting $\text{Im}\{k'_{\pm}\} = \pm \alpha_{\pm}$, Eq. (32) supplies

$$\begin{aligned} \alpha_{\pm} = & \frac{\sqrt{2}\omega}{c_0 D_p (1 \pm M)} \text{Re} \left(\sqrt{2} e^{-i\pi/4} \left[\frac{1}{(1 \pm M)^2} \sqrt{\nu} \right. \right. \\ & \times F_A \left(\sqrt{\frac{i\omega\nu}{\kappa^2 \nu_*^2}}, \delta_v \sqrt{\frac{i\omega}{\nu}} \right) \\ & \left. \left. + \frac{\beta c_0^2}{c_p} \sqrt{\chi} F_A \left(\sqrt{\frac{i\omega\chi P_t^2}{\kappa^2 \nu_*^2}}, \delta_v \sqrt{\frac{i\omega}{\chi}} \right) \right] \right). \quad (34) \end{aligned}$$

This formula suggests that the damping is strongly dependent on the mean stream Mach number M . Now mean flow convection terms of the type $U(x_2) \partial v_1 / \partial x_1 \approx i\omega v_1 U(x_2)/c_0$, etc., were neglected in deriving the low Mach number acoustic boundary layer equations (16) and (22). The validity of the present application therefore requires that such terms, which are nominally $O(M)$, should continue to be negligible at small but finite values of the pipe flow Mach number.

The mean flow causes the damping coefficients $\alpha_{\pm}$ for waves of equal frequency (propagating in the upstream and downstream directions) to be different. This is shown in Fig. 4 for sound of frequency $\omega/2\pi=88$ Hz propagating in air in a hard walled circular cylindrical pipe of radius $R=1.5$ cm. Values of $\alpha_{\pm}/\alpha_0$ predicted by (33) and (34) [δ_v being defined

as in (18), with $\sigma=1.7$] are plotted against Mach number, and compared with measurements of Peters *et al.*^{9,14} at the same frequency. For each value of $M=U/c_0$ the friction velocity v_* is determined from the empirical pipe flow formula²⁰

$$U/v_* = 2.44 \ln(v_* R/\nu) + 2.0.$$

The frequency ratio ω/ω_* becomes progressively smaller as the Mach number increases, with $\omega \approx \omega_*$ near $M=0.05$ (where the attenuations exhibit local minima) and $\omega/\omega_* \approx 0.3$ at $M=0.1$. The momentum and thermal acoustic boundary layer thicknesses also increase with Mach number, being equal to about $0.2R$ at $M=0.1$. The agreement with experiment is good, except for the discrepancy for α_-/α_0 at very low Mach numbers. The experimental results here seem to be anomalous, inasmuch as it is difficult to see how Mach numbers as low as 0.02 in air can have such a strong effect on the attenuation. Alternatively, when $M < 0.03$, the Reynolds number based on pipe radius is less than 10^4 , which may be too small to justify the high Reynolds number eddy viscosity model of Sec. II.²⁰

IV. CONCLUSION

Sound of low frequency propagating through turbulence is dissipated by the transfer of energy to turbulence kinetic and thermal energy. The transfer rates are enhanced for turbulent wall flows, where acoustic gradients become large. The analysis of this problem discussed in this paper is strictly applicable to low Mach number turbulent wall flows. Good agreement is obtained with recent data reported by Peters *et al.*^{9,14} Pipe flow measurements at *very small Mach number* yield attenuations that are marginally larger than the predictions for waves propagating against the mean stream. The discrepancy seems to be too large to be wholly explained by Mach number effects, and may arise because the Reynolds number at these Mach numbers is less than about 10^4 , which might be small enough to invalidate the eddy viscosity model of the present theory, or it may be associated with an anomaly in the measurements.

ACKNOWLEDGMENT

This work was sponsored by the Applied Hydrodynamics Research Program of the Office of Naval Research under Grant No. N00014-93-1-0111, administered by James A. Fein.

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